Market Regimes and Equity Volatility: Identifying Market States and Forecasting Future Volatility

SIADS 696 Milestone II Project Report

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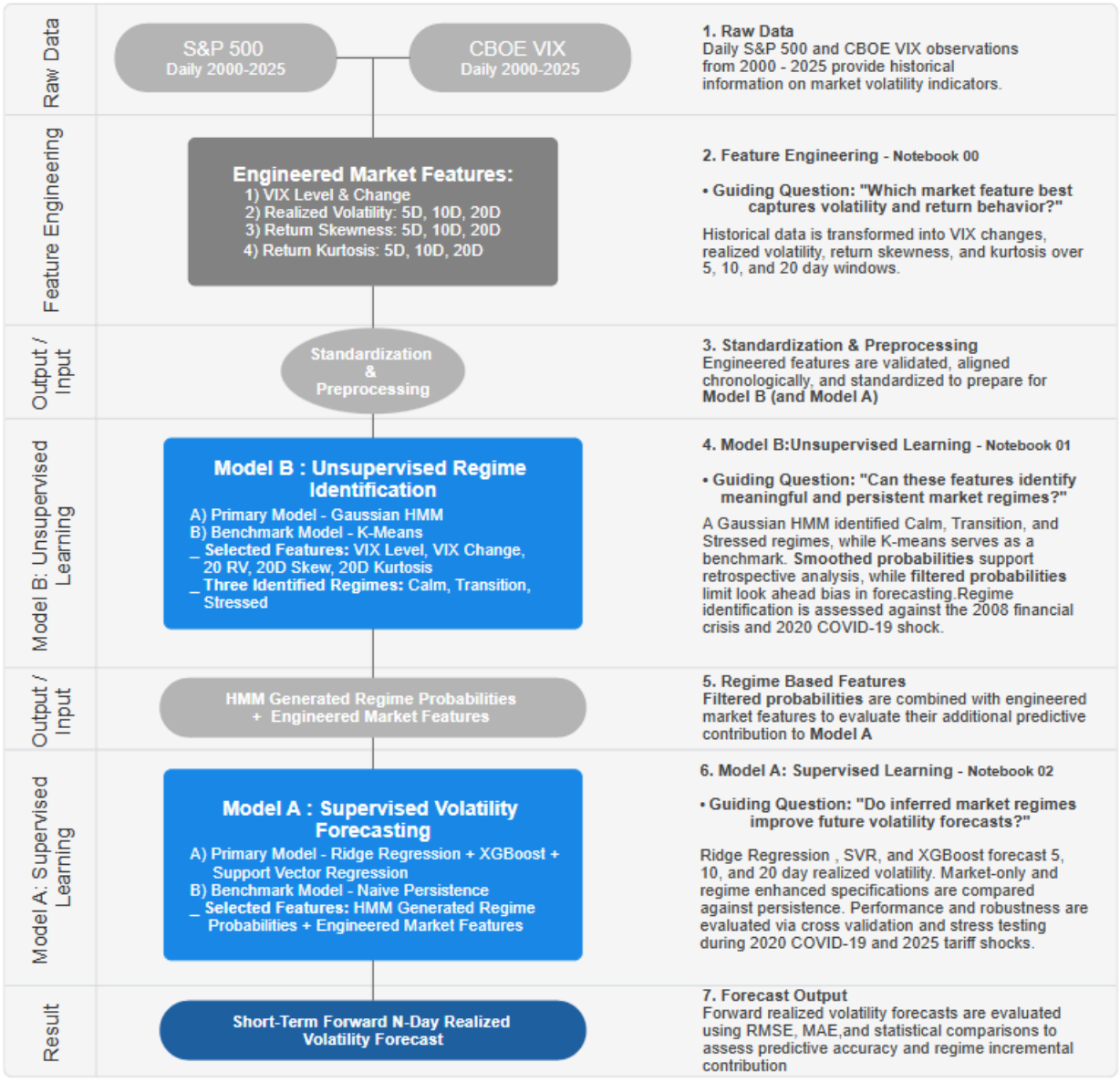
Team 22

I. Introduction

Just like the economy, financial markets experience volatility that is prone to fluctuations between periods of stability and heightened uncertainty.This project examines whether identifying distinct market regimes through unsupervised learning can identify information that can help improve forecasting future market volatility.

Using daily S&P 500 and CBOE VIX index data from 2000 to 2025, this project combines unsupervised learning regime identification (Model B) with supervised learning volatility forecasting (Model A). The workflow illustrated in Figure 1, evaluates whether the information identified pertaining to a certain market condition can improve predictive value beyond conventional market indicators.

Figure1. Integrated Machine Learning Workflow for Market Regime Identification and Volatility Forecasting



II. Related Work

The project builds on four main studies supporting it's framework:

Model B (Unsupervised Learning): Ang and Bekaert (2002) establish the economic relevance of market regime shifts, while Wang et al. (2020) demonstrate the use of Hidden Markov Models (HMM's) for regime identification.

Feature Engineering and Model A (Supervised Learning): Corsi (2009) motivates the use of multi-horizon volatility features, while Zhuo and Morimoto (2024) defend Support Vector Regression (SVR) as a nonlinear approach to realized volatility forecasting.

Research contribution is derived from integrating both regime identification with volatility forecasting to evaluate whether HMM learned probabilities improve predictive accuracy beyond conventional market indicators.

III. Data Sources and Feature Engineering

3.1 Data Sources and Preparation

Daily S&P 500 Index and CBOE Volatility Index (VIX) data were obtained from Yahoo Finance, covering January 2000 to December 2025. The original dataset contained 6,539 and 6,567 observations respectively, with no missing values or duplicate dates. An inner join on trading dates excluded 28 unmatched VIX observations (markets were closed) , producing a consolidated dataset of 6,539 observations and 11 variables, stored in paraquet format.

3.2 Feature Engineering

Following data integration, Notebook 00 generates market indicators capturing implied volatility, historical volatility, and return distribution characteristics. Daily S&P 500 logarithmic returns are calculated and used to derive rolling realized volatility, skewness, and kurtosis over 5, 10, and 20 day rolling windows. While VIX closing values and their daily changes were used to capture market sentiment towards uncertainty

Table 1: Engineered Market Features and Descriptions in Accordance with Model Utilization

Feature	Description	Model
VIX_Level	VIX closing value, reflecting expected near term market volatility.	A & B
VIX_Change	Day over day change in the VIX capturing shifts in market uncertainty.	A & B
Realized_Vol_5D/10D/20D	Trailing volatility estimated from daily S&P 500 returns over 5, 10, and 20 trading days.	A,(20D for B)
Return_Skew_5D/10D/20D	Rolling skewness, measuring asymmetry in historical return distributions.	A,(20D for B)
Return_Kurt_5D/10D/20D	Rolling kurtosis, describing the tail characteristics of historical returns.	A,(20D for B)
Filtered_P_Calm	Filtered HMM probability of the Calm regime	A*
Filtered_P_Transition	Filtered HMM probability of the Transition regime	A*
Filtered_P_Stressed	Filtered HMM probability of the Stressed regime	A*

Note*: Model B uses five selected market features. Model A uses eleven market based predictors, supplemented by three filtered regime probabilities in the regime enhanced specification. Model A forecasts forward volatility over 5, 10, and 20 trading days.

3.3 Preprocessing and Model Integration

Engineered features were validated for completeness and chronological consistency before export in paraquet format. Cutoffs were introduced throughout Model A and Model B to **limit the influence of data leakage.** Model B separated training (2000 - 2017) and testing (2018 - 2025) periods, while Model A used training (2000 - 2018), validation (2019- 2021), and testing (2022- 2025), along with a 20 day purge at the partition boundaries. Added feature standardization used training period statistics to further limit the influence.

Model B generated two sets of probabilities: smoothed probabilities that included past and future observations for retrospective regime analysis, and filtered probabilities, which use information available up to each trading date to **limit look ahead bias**. The filtered probabilities were used by Model A to determine whether regime information contributed anything to forecasting value.

IV. Model B: Unsupervised Regime Identification

4.1 Objective and Methodology

Model B addressed the research question : **Can unsupervised learning identify economically interpretable and persistent market regimes in S&P 500 and VIX data, and do these regimes remain meaningful outside the training period?**

A Gaussian Hidden Markov Model (HMM) was selected to capture temporal dependence and regime transitions, K-means serving as a non-temporally sensitive clustering benchmark. Five standardized market features were used, focusing on 20 day window features to capture regime persistence and reduce sensitivity to short term fluctuations. Observations were chronologically divided into training (2000-2017) and testing (2018- 2025)

The analysis addressed the research question through four stages:

11. **Regime identification:** HMM specifications with 2-6 states were evaluated using Bayesian Information Criterion (BIC), sensitivity analysis, and economic interpretability to identify three regimes: Calm, Transition, and Stressed.
12. **Economic relevance and stability:** Regime characteristics were compared across training and testing periods, using the 2008 Global Financial Crisis and 2020 COVID-19 market shocks as historical benchmarks to assess whether regimes aligned with periods of financial turmoil.
13. **Persistence and benchmarking:** Transition probabilities, expected regime durations, and comparisons with K-means evaluated clustering performance and regime duration.
14. **Model integration:** Smoothed probabilities supported retrospective investigations regarding uncertainty surrounding regime transitions and classification confidence. Filtered probabilities were generated for Model A to limit look ahead bias.

4.2 Model Selection and Regime Identification

Gaussian HMM specifications with 2–6 hidden states were evaluated using the Bayesian Information Criterion (BIC). Although BIC favored six states, sensitivity analysis revealed a tradeoff between statistical fit and regime persistence, switching became more frequent as states increased. A three state model was selected for its economic interpretability and clearer distinction between Calm, Transition, and Stressed conditions. Balancing complexity, economic interpretability, and temporal stability.

4.3 Regime Characteristics and Economic Validation

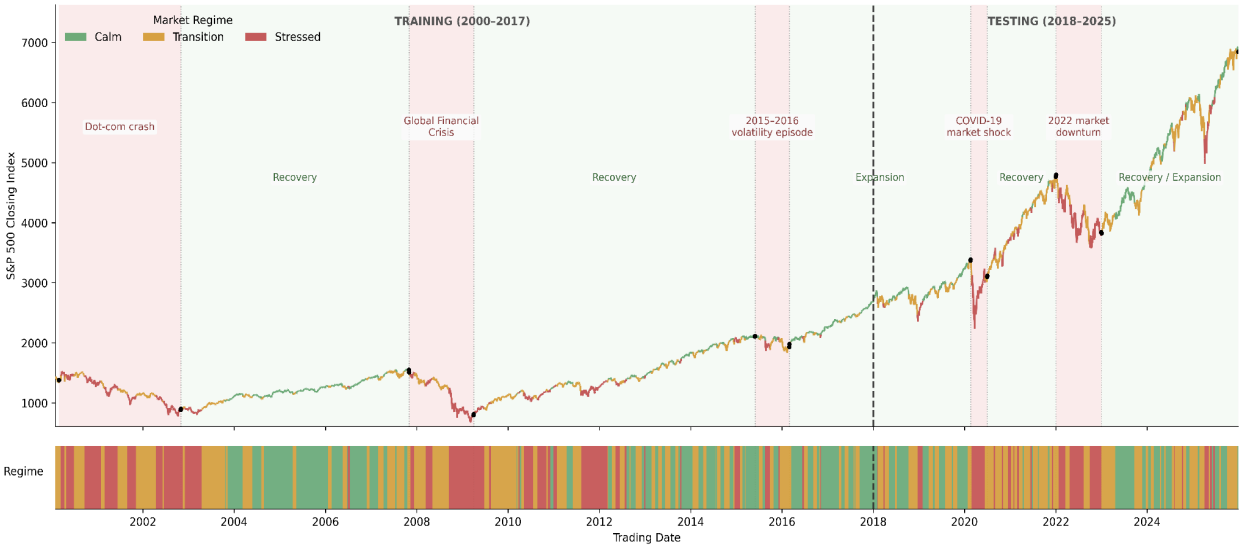
The three state model identified Calm, Transition, and Stressed market conditions, as shown in Table 2, Calm exhibited the lowest VIX levels and realized volatility, Transition intermediate volatility with negative return skewness and elevated kurtosis, and Stressed the highest volatility.

Table 2: HMM Regime Characteristics Across Training and Testing Periods

Regime	Period	Avg. VIX	Avg Realized Vol. (%)	Avg Return Skewness	Avg Return Kurtosis
Calm	Training	**13.4101**	9.3080	0.2008	0.3846
Calm	Testing	**14.2401**	9.6354	0.0933	0.1115
Transition	Training	19.2329	15.4896	**-0.4937**	**0.6472**
Transition	Testing	19.5088	15.7165	**-0.5477**	**0.6514**
Stressed	Training	**29.4197**	27.0164	0.2846	0.3363
Stressed	Testing	**28.8154**	28.2890	0.1699	0.6015

Note: Realized volatility represents the annualized 20 day measure. Training observations cover 2000 - 2017, while testing observations cover 2018 - 2025.

Figure 2: S&P 500 Market Regimes Identified by the Three State HMM (Training and Testing Periods: 2000 - 2025)



In and out of sample regime assignments revealed a preference towards Transition conditions, increasing from 38% during training to 49.13% during testing. Calm and Stressed assignment decreased by 4.76% and 6.37% respectively. Despite these changes, regimes retained broadly consistent economic characteristics, supporting their out of sample interpretability.

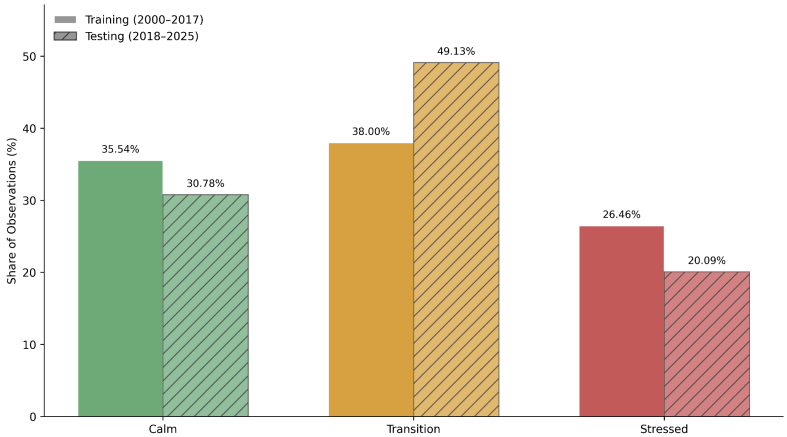


Figure 3: HMM Regime Prevalence: Training (2000 - 2017) vs. Testing (2018-2025)

4.4 Regime Persistence and Model Comparison

The HMM demonstrated strong regime persistence, with self-transition probabilities of 0.9505–0.9749 and implied durations of 20–40 trading days (Table 3). Persistence weakened during testing across all regimes, particularly Stressed, whose expected duration declined from 39.77 to 20.20 days.

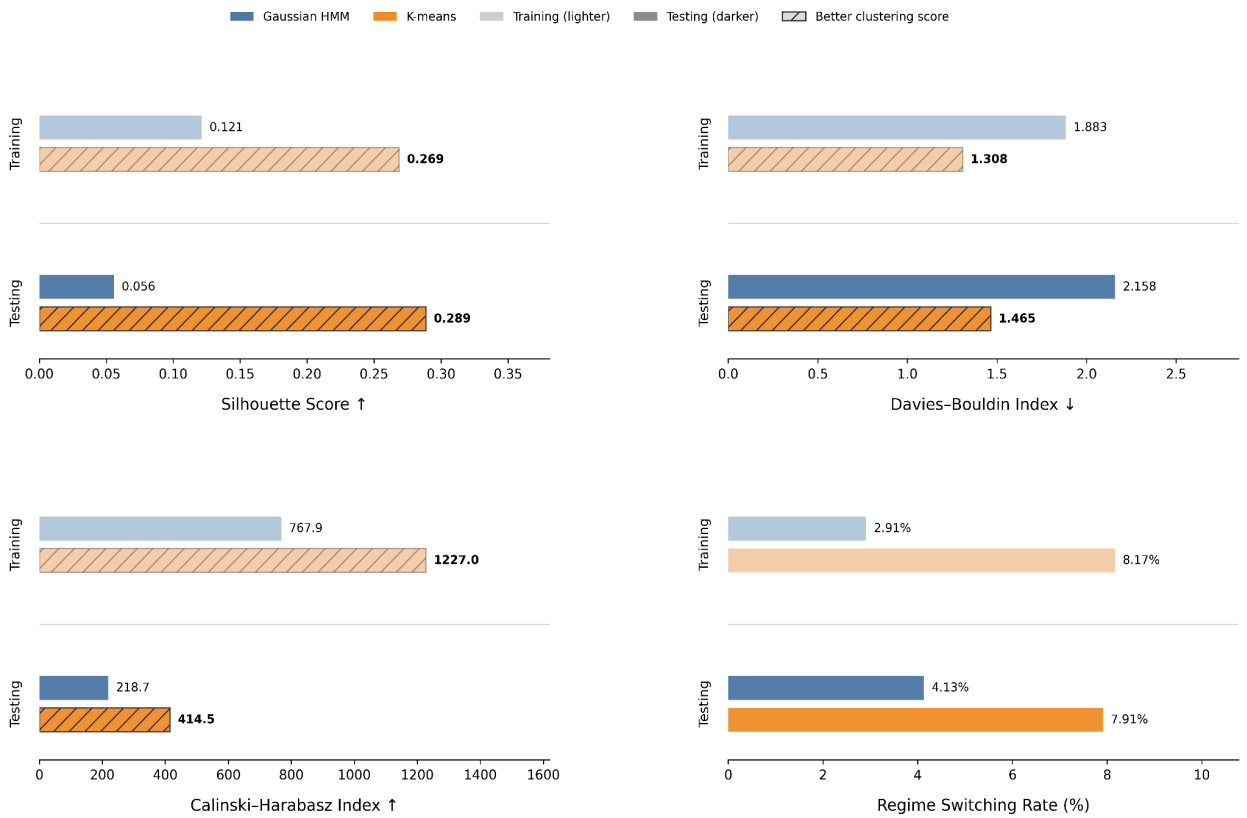
Table 3: HMM Regime Persistence: Training vs. Testing.

Regime	Period	Self – Transition Probability	Exit Probability	Implied Duration (Days)
Calm	Training	0.9725	0.0275	36.39
Calm	Testing	0.9612	0.0388	25.75
Transition	Training	0.9667	0.0333	30.05
Transition	Testing	0.9605	0.0395	25.33
Stressed	Training	0.9749	0.0251	39.77
Stressed	Testing	0.9505	0.0495	20.20

Note: Transition probabilities and implied durations are derived from retrospectively HMM regime assignments (not forecasted) . Training covers 2000–2017; testing covers 2018–2025.

K-means performed better in cluster separation, reflected in the silhouette score and davies bouldin index in both training and testing (Figure 4). However, the HMM produced fewer regime switches, this could mean that market conditions changed more frequently. These results highlight the tradeoff between cluster separation and temporal stability. The HMM was preferred because its ability to capture persistent market conditions better aligns with the objective of identifying market regimes.

Figure 4:HMM vs. K-means: Clustering Performance and Regime Stability (Training and Testing)



4.5 Regime Probabilities and Model A Integration

Smoothed HMM probabilities were used to assess classification confidence and uncertainty around regime transitions, particularly during the 2008 financial crisis and 2020 COVID-19 shock. Filtered probabilities were generated using information available up to each trading date to limit look ahead bias and integrated into Model A to evaluate their contribution to volatility forecasting.

V. Model A: Supervised Volatility Forecasting

5.1 Objective and Methodology

Model A addresses the research question:**Can supervised learning forecast future S&P 500 realized volatility, and does incorporating HMM inferred market regime information improve forecasting performance beyond conventional market indicators?**

Ridge Regression, Support Vector Regression (SVR), and XGBoost were evaluated against a naive persistence benchmark to forecast realized volatility over 5, 10, and 20 trading days. Each model was tested using market-only and regime enhanced predictors. Data was split into training (2000–2018), validation (2019–2021), and testing (2022–2025), with a 20-day purge at partition boundaries to limit data leakage.

The analysis addressed the research question through four stages:

1. **Forecast performance:** RMSE and MAE assessed predictive accuracy against the persistence benchmark across all three horizons.
2. **Incremental HMM contribution:** Market-only and regime enhanced models were compared to measure the additional forecasting value of regime probabilities.
3. **Model robustness:** Expanding window cross-validation, hyperparameter sensitivity, and feature ablation assessed forecasting stability and predictor probabilities.
4. **Failure analysis:** Forecast errors were examined during periods of market instability, including the 2020 COVID-19 and 2025 tariff shocks.

5.2 Forecasting Performance

All three supervised models outperformed the naive persistence benchmark across all forecasting horizons (Table 4). Market-only Ridge Regression consistently achieved the best performance, followed by SVR, while HMM-enhanced XGBoost performed worst among the supervised models. Adding HMM probabilities generally reduced forecasting accuracy, with SVR showing a slight improvement at the 5-day horizon. Overall, the findings suggest that conventional market indicators provided stronger predictive value than the additional regime information.

Table 4: Out of Sample RMSE and Improvement over Persistence Benchmark in Percentage.

Forecast Horizon	Persistence	Ridge (Market-only)	Ridge (+ HMM)	SVR (Market-only)	SVR (+ HMM)	XGBoost (Market-only)	XGBoost (+ HMM)
5-Day	0.0974	**0.0738 (24.23%)**	**0.0740 (24.05%)**	0.0744	0.0741	0.0756	0.0761 (21.885%)
10-Day	0.0849	**0.0670 (21.11%)**	**0.0673 (20.67%)**	0.0694	0.0699	0.0697	0.0704 (17.093%)
20-Day	0.0761	**0.0607 (20.22%)**	**0.0611 (19.69%)**	0.0626	0.0626	0.0642	0.0645 (15.254%)

Note: Results cover the 2022- 2025 testing period. Bold indicated the best supervised model , underlined indicated the worst, Percentages represent RMSE reductions relative to persistence.

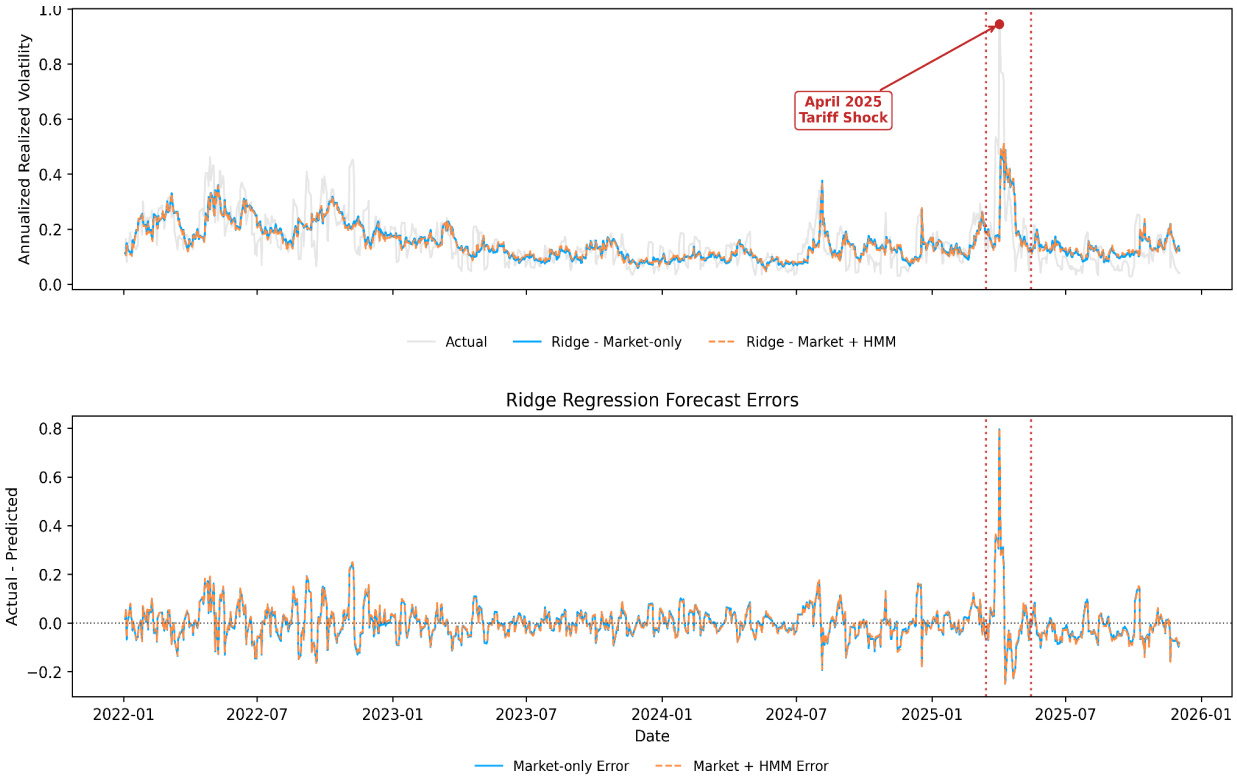
Table 5: Out of Sample Forecasting Performance (MAE).

Forecast Horizon	Persistence	Ridge (Market-only)	Ridge (+ HMM)	SVR (Market-only)	SVR (+ HMM)	XGBoost (Market-only)	XGBoost (+ HMM)
5-Day	0.0647	0.0492	0.0494	0.0492	0.0489	0.0497	0.0500
10-Day	0.0543	**0.0427**	0.0428	0.0429	0.0432	0.0428	0.0428
20-Day	0.0487	**0.0392**	0.0393	0.0395	0.0396	0.0398	0.0399

Note: Bold indicates the lowest supervised MAE, underlined indicated the highest supervised MAE. Several models tie at the displayed precision.

The RMSE and MAE scores indicate that prediction errors remain, even for the best performing models. These errors include underestimating the magnitude of future volatility, particularly during sudden market disruptions (Figure 5). This also highlights a limitation of the data, as historical S&P 500 and VIX indicators may not capture all factors driving unexpected changes in market conditions.

Figure 5: Ridge Regression: 5-Day Volatility Forecasting (Actual vs Predicted).

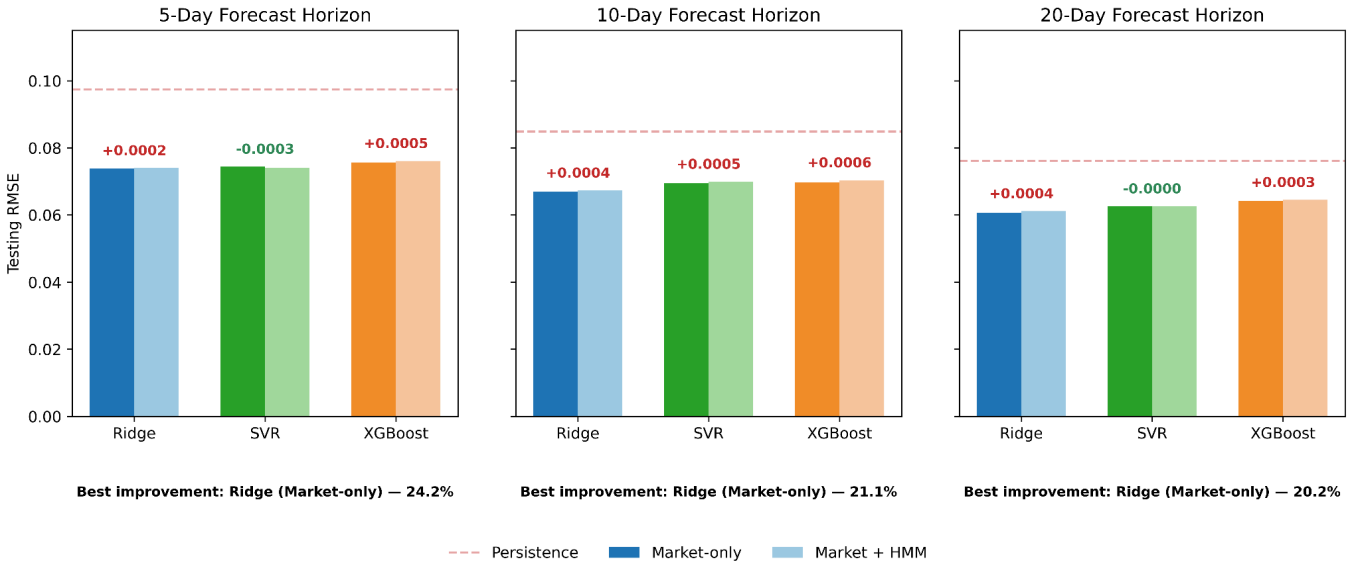


5.3 Incremental Value of HMM Regime Information

Adding HMM probabilities generally reduced forecasting accuracy, except for a slight improvement in SVR at the 5-day horizon. XGBoost experienced the greatest deterioration, while forecast errors increased with longer forecasting horizons.One possible explanation is that HMM probabilities capture

market conditions already reflected in VIX levels and historical realized volatility, resulting in overlapping information rather than improved forecasts.

Figure 6: Change in Out-of-Sample Forecasting RMSE from Adding HMM Regime Probabilities



5.4 Robustness and Sensitivity Analysis

The expanding window cross-validation analysis used four folds: (I) 2012–2014, (II) 2015–2017, (III) 2018–2019, and (IV) 2020–2021, with training data expanding across successive folds and a 20 day purge to limit leakage. Ridge Regression performed most consistently, ranking first in 8 of 12 comparisons and achieving an average RMSE improvement of 19.1985% over persistence. However, forecasting errors increased during the COVID-19 period (Fold IV, Figure 7). Sensitivity analysis showed Ridge was relatively stable, while feature ablation identified VIX indicators and historical volatility as the strongest predictors, with HMM probabilities adding limited value.

Figure 7: Expanding Window Cross-Validation: RMSE Across Historical Folds

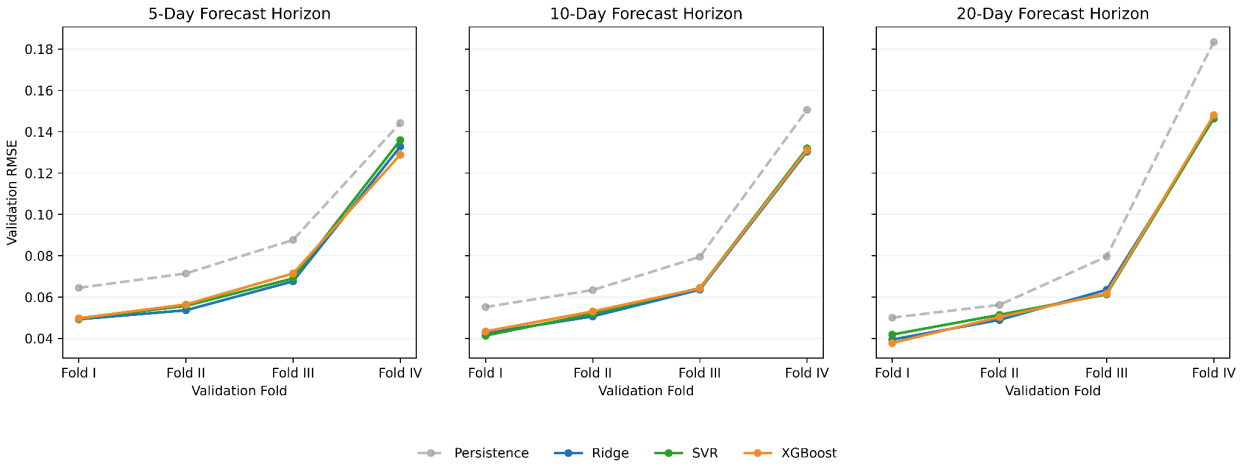
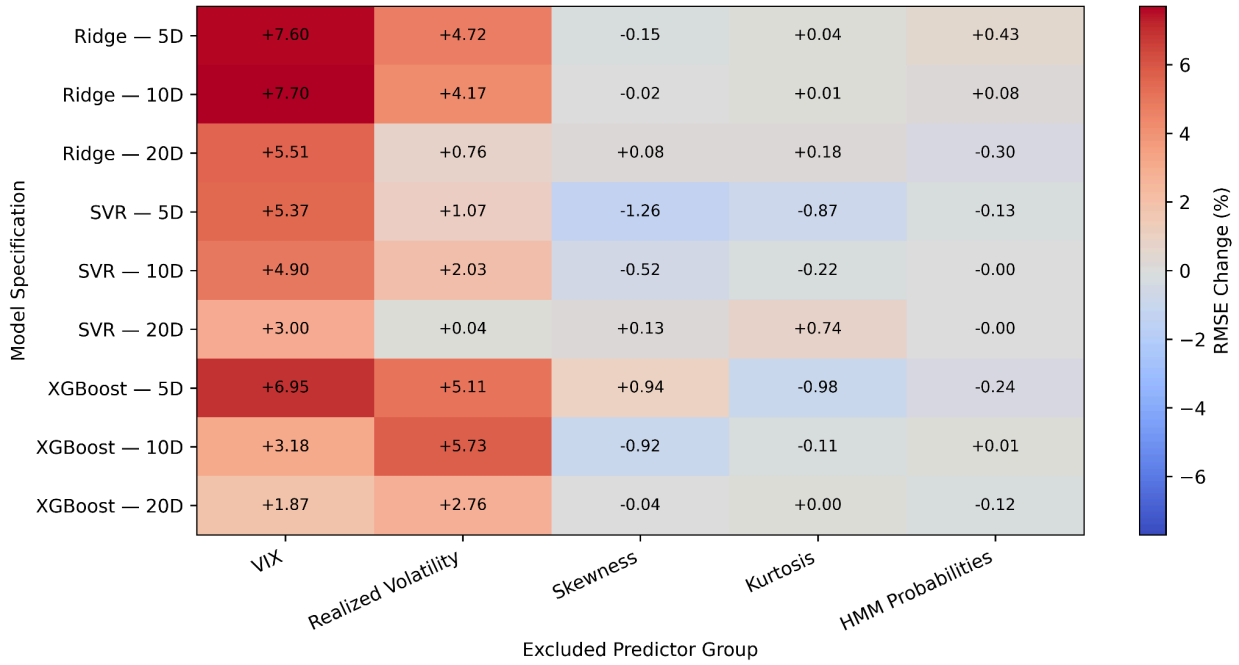


Figure 8: Feature Ablation : Change in Validation RMSE (%)



Feature ablation confirmed that VIX indicators and historical realized volatility were the strongest predictors, with their removal increasing validation RMSE by 5.12% and 2.93% on average, respectively. In contrast, removing skewness, kurtosis, and HMM probabilities slightly improved or had minimal impact on forecasting accuracy, suggesting limited additional predictive value from these features.

5.5 Stress Period Evaluation and Failure Analysis

Table 6: Forecasting Performance During Market Stress

Market Event	Model	Stress RMSE	Non-Stress RMSE	RMSE Increase (%)	Stress MAE
COVID-19 (2020)	Ridge	0.2667	0.0756	252.8406%	0.1737
	SVR	0.2778	0.0734	278.2388%	0.1869
	XGBoost	**0.2572**	0.0746	**244.9206%**	0.1739
Tariff Shock (2025)	**Ridge**	**0.2149**	0.0599	**258.9766%**	**0.1387**
	SVR	0.2186	0.0600	264.2978%	0.1430
	XGBoost	0.2285	0.0599	281.1633%	0.1595

Note: Stress periods refer to heightened market volatility during the COVID-19 shock (February 20–June 30, 2020) and the 2025 tariff related disruption (March 15–May 15, 2025). Non-stress periods refer to observations outside these event windows within the corresponding evaluation samples. Results are based on market-only 5 day forecasts.

Forecasting errors increased sharply during both market shocks compared with non-stress periods (Table 6). XGBoost achieved the lowest RMSE during COVID-19, while Ridge performed best during the 2025 tariff shock. These findings suggest that model performance varies across market conditions, with all three models struggling to anticipate sudden volatility increases.

VI. Model B and Model A Concluding Remarks

6.1 Interpretation of Findings

The findings highlight a distinction between identifying market regimes and improving volatility forecasts. **Model B** successfully identified three economically meaningful market conditions: Calm, Transition, and Stressed. These conditions remained distinguishable outside the training sample, although regime persistence weakened during testing.

Model A demonstrated that conventional market indicators were more effective for forecasting future volatility. Ridge Regression achieved the strongest overall performance, while HMM derived probabilities generally reduced forecasting accuracy. This suggests that regime information may capture patterns already reflected in VIX levels and historical realized volatility, offering limited additional predictive value.

Overall, identifying meaningful market regimes does not necessarily translate into more accurate volatility forecasts.

6.2 Model Limitations

The analysis relied exclusively on S&P 500 and VIX data, excluding other drivers of volatility, such as macroeconomic developments and market liquidity. Additionally, changes to the VIX methodology in 2003 and 2014 may affect consistency across historical observations.The HMM's assumption of three market states may oversimplify changing market conditions, while fixed training periods limit model adaptability. Furthermore, the supervised models struggled to anticipate sudden volatility spikes, highlighting the limitations of forecasting unexpected market disruptions using historical indicators alone.

6.3 Implications and Future Research

The findings suggest that market regimes are useful for understanding market conditions but offer limited additional value in forecasting volatility. A key limitation is the underrepresentation of broader global uncertainty, including geopolitical tensions, economic disruptions, and policy changes. Future research could construct a range of proxy indicators for these uncertainties to better capture their influence on market volatility and potentially improve forecasting performance.

6.4 Ethical Considerations

Because volatility forecasts can influence financial decisions, it is important to communicate the model's limitations clearly. The models struggled to predict sudden market shocks, and adding HMM regime information generally did not improve forecasting accuracy. Being transparent about these findings helps prevent misleading conclusions. Forecasts should therefore be viewed as useful analytical tools rather than reliable predictions or financial advice.

VII. Conclusion

This study addressed two main research questions: **Can unsupervised learning identify meaningful and persistent market regimes? And does incorporating these regimes improve forecasts of future equity market volatility?**

The Gaussian HMM successfully identified three distinct market regimes: Calm, Transition, and Stressed, addressing the first question. However, incorporating regime probabilities generally did not improve forecasting accuracy, with Ridge Regression achieving the strongest overall performance. These findings suggest that while market regimes help explain changing market conditions, predicting future volatility remains challenging, particularly during unexpected market disruptions.

VIII. Statement of Work

Statement contributions and deliverables (Available on Git Repository - including research cited)

The project involved literature review, data preparation, feature engineering, unsupervised and supervised model development, model evaluation, visualization, and report preparation. All notebooks, supporting visualizations, and analytical outputs are available in the project GitHub repository.

Member	Contribution	Deliverable
Lina Al-Rawahi	Literature review and research framework development	References
	Market data validation, feature engineering, feature selection, and dataset preparation	Notebook 00: Feature Engineering
	HMM development, regime identification, model selection, and K-means benchmarking	Notebook 01: Model B – Market Regime Identification
	Regime stability, transition dynamics, sensitivity analysis, and historical market diagnostics	Notebook 01: Model B – Market Regime Identification
	Forward volatility target construction, supervised model development, and hyperparameter tuning	Notebook 02: Model A – Supervised Volatility Forecasting
	Forecast evaluation, expanding window cross-validation, feature ablation, and market stress analysis	Notebook 02: Model A – Supervised Volatility Forecasting
	Development and export of analytical visualizations and modeling outputs	Model A and Model B Figures and Outputs
	Integration of research findings, report writing, and technical documentation	Milestone II Project Report and Appendices
	Project organizing and GitHub repository management	Project repository, standardized notebooks
Generative AI Disclosure: Generative AI tools supported in development, proofreading, and organizing activities.		

References

Ang, A., & Bekaert, G. (2002). International asset allocation with regime shifts. The Review of Financial Studies, 15(4), 1137–1187.

Corsi, F. (2009). A simple approximate long-memory model of realized volatility. Journal of Financial Econometrics, 7(2), 174–196.

Wang, M., Lin, Y.-H., & Mikhelson, I. (2020). Regime-switching factor investing with hidden Markov models. Journal of Risk and Financial Management, 13(12), 311.

Zhuo, Y., & Morimoto, T. (2024). A hybrid model for forecasting realized volatility based on heterogeneous autoregressive model and support vector regression. Risks, 12(1), 12.

Literature Review Matrix - Conceptual Foundation and Contribution to Project Deliverables

Research	Reference	Conceptual Contribution	Project Deliverable

Theme			
Market Regime Theory & Identification	Ang & Bekaert (2002)	Demonstrates the importance of regime dependent volatility and correlations in financial markets	**Model B** : Supports the economic rationale for identifying and interpreting distinct market conditions
	Wang et al. (2020)	Demonstrates how HMMs can identify latent market regimes and support investment strategies	**Model B :** Supports the choice of an HMM for classifying latent market states
Volatility Dynamics	Corsi (2009)	Explains volatility persistence and the importance of historical volatility over multiple time horizons	**Feature Engineering :** Used trailing 5, 10, and 20 day realized volatility features.
Volatility Forecasting	Zhuo & Morimoto (2024)	Examines the use of Support Vector Regression (SVR) for forecasting realized volatility and capturing non-linear volatility relationships	**Model A:** Support the selection of SVR as a nonlinear forecasting approach alongside Ridge Regression and XGBoost

Appendices

Appendix A : Data Dictionary and Feature Definitions

Dataset Description and Dimensions

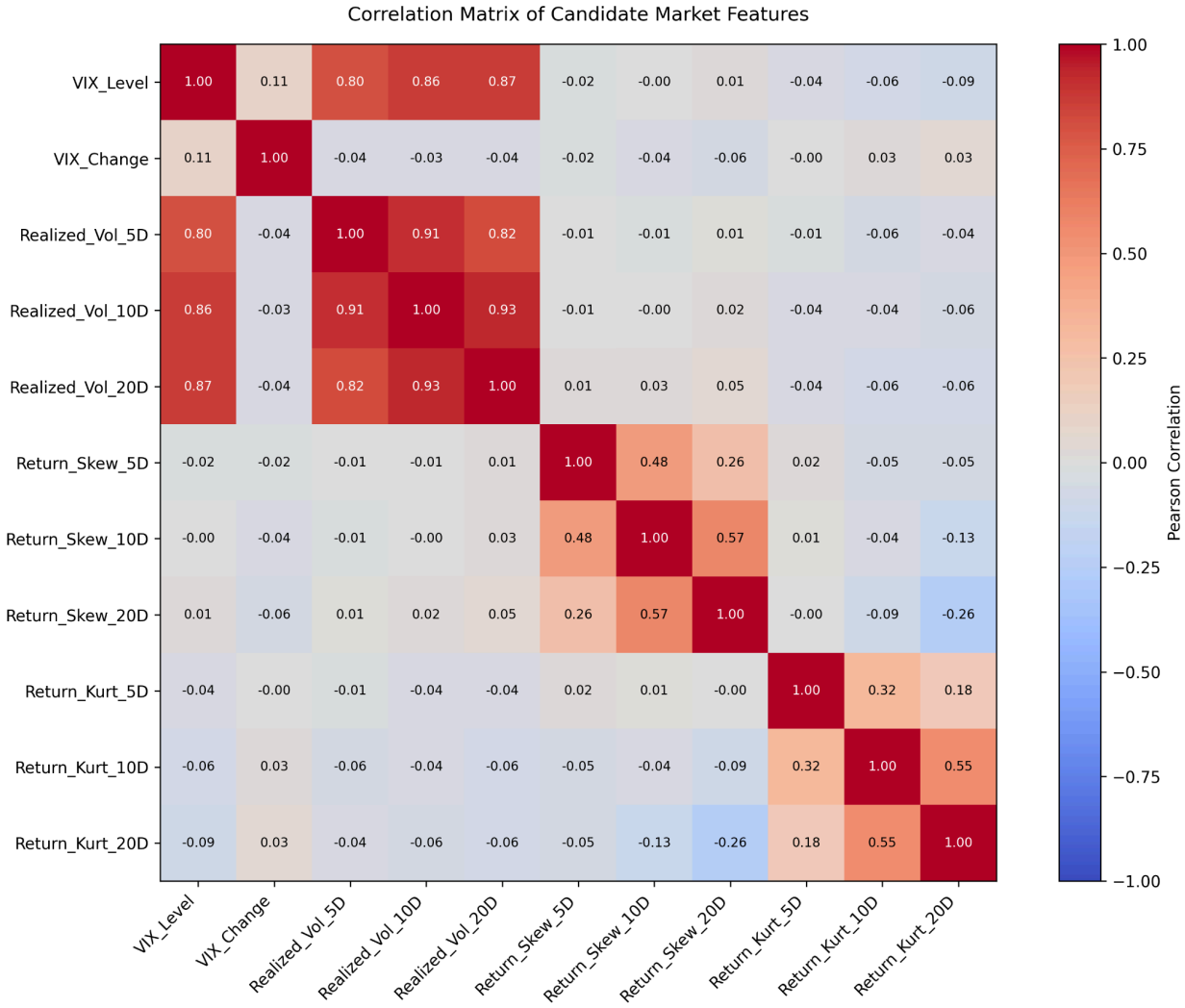
1. **Original Market Data (6,539 × 11):** Consolidated daily S&P 500 and CBOE VIX observations covering January 2000 to December 2025, containing trading dates, market prices, and trading volume.
2. **Full Engineered Dataset (6,519 × 23):** Original market data supplemented by 12 engineered variables capturing logarithmic returns, implied volatility, historical realized volatility, skewness, and kurtosis over multiple trading-day windows.
3. **Selected HMM Dataset (6,519 × 7):** Subset containing five selected market indicators, trading dates, and S&P 500 closing prices, used for unsupervised regime identification in Model B.
4. **HMM Model Outputs (6,519 × 15)**: Model B dataset incorporating regime classifications, smoothed and filtered regime probabilities, and classification confidence measures alongside selected market features.
5. **Model A Integrated Dataset (6,519 × 26):** Full engineered dataset combined with three filtered HMM regime probabilities (Calm, Transition, and Stressed), providing the market-based and regime-enhanced predictors used in supervised volatility forecasting.

Consolidated Data Dictionary and Model Utilization

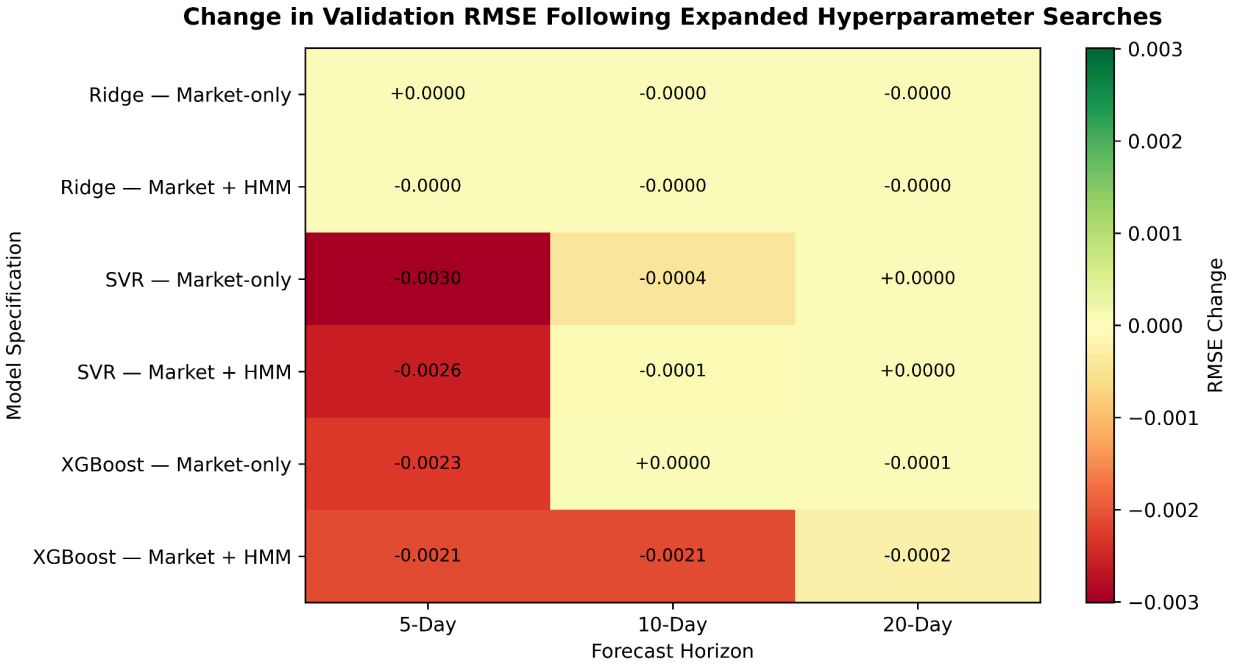
Category	Variable	Description	Model
Original Data	Date	Trading date used to align market observations.	A & B
	SP500_Open	S&P 500 opening index value.	—
	SP500_High	Highest S&P 500 index value during the session.	—
	SP500_Low	Lowest S&P 500 index value during the session.	—
	SP500_Close	Closing index value used to calculate historical returns.	A & B
	SP500_Adj_Close	Adjusted S&P 500 closing value.	—
	SP500_Volume	Reported daily trading volume.	—
	VIX_Open	Opening VIX index value.	—
	VIX_High	Highest VIX index value during the session.	—
	VIX_Low	Lowest VIX index value during the session.	—

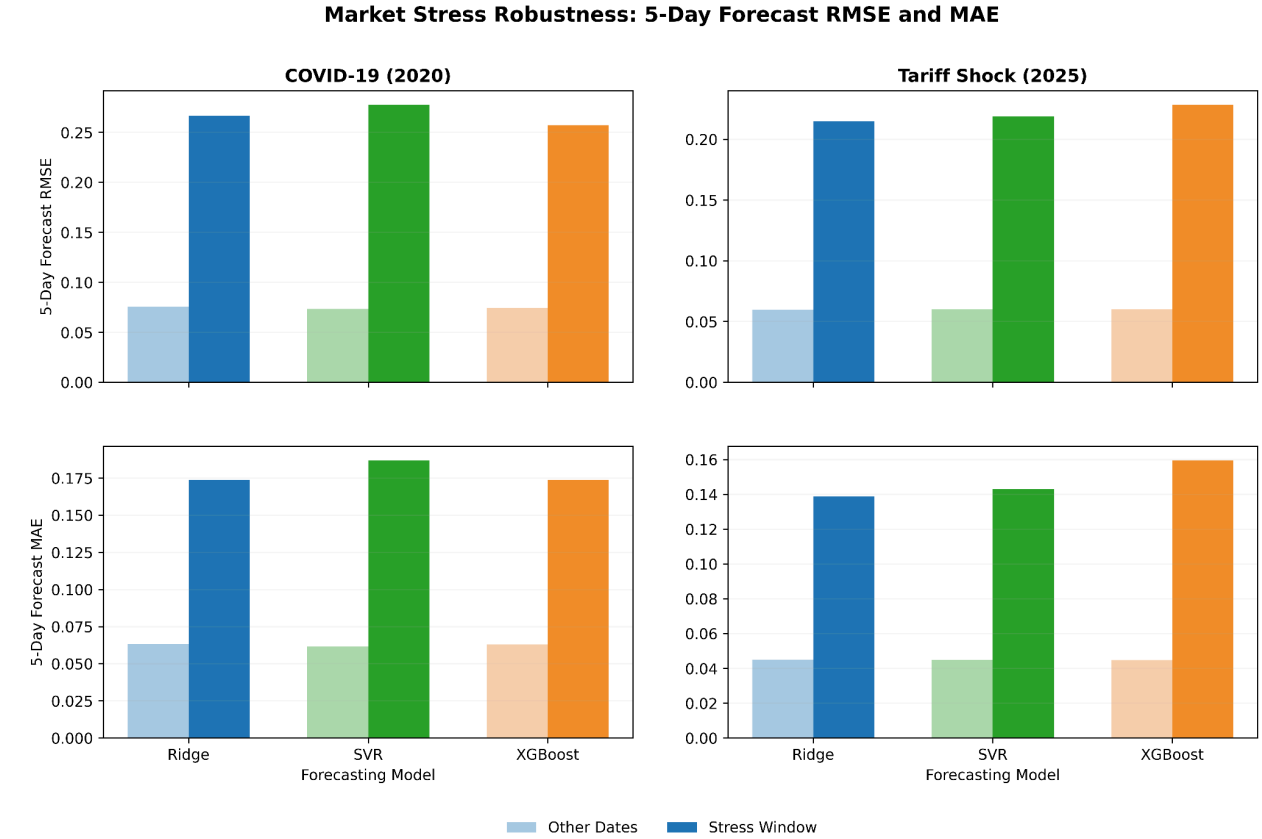
	VIX_Close	Closing VIX value used to construct implied volatility indicators.	A & B
Engineered Features	SP500_Log_Return	Daily logarithmic return calculated from consecutive S&P 500 closing values.	Intermediate
	VIX_Level	VIX closing value representing implied market volatility.	A & B
	VIX_Change	Day-over-day change in VIX closing values.	A & B
	Realized_Vol_5D/10D/20D	Annualized standard deviation of trailing logarithmic returns over 5, 10, and 20 trading days.	A; 20D for B
	Return_Skew_5D/10D/20D	Rolling skewness measuring asymmetry in historical return distributions.	A; 20D for B
	Return_Kurt_5D/10D/20D	Rolling excess kurtosis measuring historical return distribution tail characteristics.	A; 20D for B
Forecast Targets	Forward_RV_5D/10D/20D	Annualized realized volatility over the following 5, 10, and 20 trading days.	A
HMM Outputs	Filtered_P_Calm	Estimated probability of the Calm regime using information available through the current date.	B → A
	Filtered_P_Transition	Estimated probability of the Transition regime using information available through the current date.	B → A
	Filtered_P_Stressed	Estimated probability of the Stressed regime using information available through the current date.	B → A

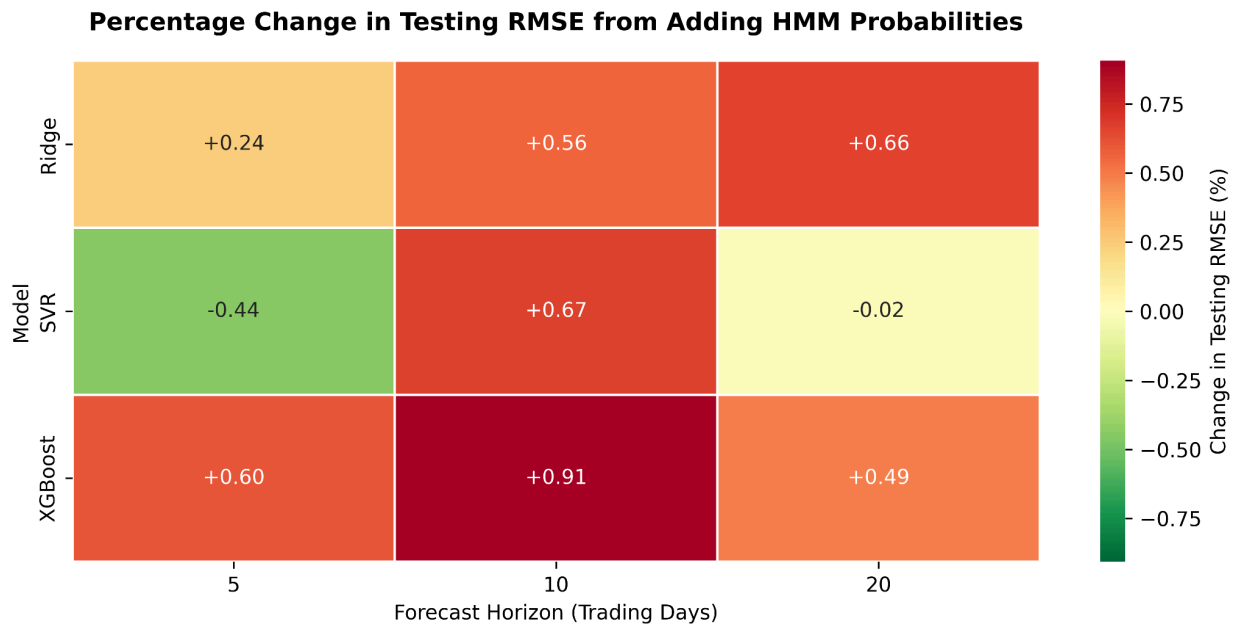
Appendix B : Feature Engineering - Supplementary Visualizations



Appendix C : Model A - Supplementary Visualizations






Percentage Change in Testing RMSE from Adding HMM Probabilities
Model
Ridge
SVR
XGBoost
+0.24
+0.56
+0.66
-0.44
+0.67
-0.02
+0.60
+0.91
+0.49
5
10
20
Forecast Horizon (Trading Days)
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
Change in Testing RMSE (%)